Attention, Salience, Complexity, and Learning

Behavioral Labor – Week 4

Teaching deck draft

Central question

- How do attention, salience, complexity, learning, and costly information acquisition shape labor choices?
- Week 3 studied beliefs and expectations in job search.
- Week 4 asks what happens when the relevant labor rule exists but is hard to notice, understand, or update.
- Core margins: earnings, hours, take-up, claiming, effort, saving, training.

Bridge from Week 3

Week 3 object	Week 4 shift
Subjective beliefs	Workers must notice and process relevant signals
Job search information	Broader labor schedules, contracts, benefits, and programs
Perceived returns	Perceived tax, benefit, bonus, saving, and training maps
Updating over unemployment	Learning from repeated payroll, policy, and workplace exposure

DellaVigna taxonomy: Week 4 highlighted

Branch	Labor interpretation
Nonstandard preferences	Present bias, reference points, reciprocity, fairness
Nonstandard beliefs	Subjective expectations about wages, job finding, returns
Nonstandard decision-making	**Attention, salience, complexity, limited consideration, learning**
Responses	Firms and policies redesign the information environment

This week: workers respond to the schedule they perceive and learn, not only the schedule written down.

Benchmark transparent labor choice

$$a_i^\star \in \arg\max_{a\in\mathcal{A}} U_i(a;\theta_i) \quad \text{s.t.} \quad c_i = y_i + w_i a - T_i(a)$$

Object	Interpretation
a	hours, earnings, effort, claiming, saving, or training
$T_i(a)$	true tax-benefit, payroll, or contract schedule
U_i	payoff from consumption, work, effort, time, or program participation
Assumption	the worker understands the local action-to-payoff map

Salience wedge: perceived incentives

$$a_i \in \arg\max_{a \in \mathcal{A}} U_i(a; \theta_i) \quad \text{s.t.} \quad \tilde{T}_i(a) \neq T(a)$$

- The worker responds to the perceived schedule $\tilde{T}_i(a)$.
- Low salience: the relevant incentive is not prominent at the moment of choice.
- Misperception: the worker holds the wrong local slope, threshold, or eligibility rule.
- Heuristic response: average incentives replace marginal incentives.

Salience versus complexity versus information

Object	Distortion	Labor signature
Low salience	incentive exists but is not prominent	stronger response when highlighted
Complexity or opacity	action-to-payoff map is hard to decode	mistakes persist despite disclosure
Missing information	worker lacks a rule or fact	response to letters or tutorials
Endogenous information	worker chooses attention effort	response depends on stakes and costs
Dynamic learning	representation evolves with exposure	wedges shrink or change over time

Do not infer attention from attenuation alone. Name the margin and the behavioral object.

Dynamic learning and costly attention

$$a_{it}, m_{it} \in \arg\max \tilde{U}_{it}(a; \theta_i, b_{it}) - C(m_{it}) \quad \text{with} \quad b_{i,t+1} = B(b_{it}, m_{it}, x_{it+1})$$

- m_{it} is costly information acquisition or attention effort.
- b_{it} is the worker's representation of the schedule or contract.
- x_{it+1} includes experience, letters, payroll feedback, peers, and caseworkers.
- Policy and firms can change current salience and the speed of learning.

Labor supply under nonlinear schedules

- EITC, taxes, benefits, and earnings tests create nonlinear local incentives.
- Observed margins: earnings, hours, employment, bunching, and adjustment timing.
- Chetty-Friedman-Saez: local knowledge and EITC earnings responses.
- Kostol-Myhre: labor supply responses to learning the tax and benefit schedule.
- Main caution: muted bunching can reflect information, adjustment costs, constraints, or true low elasticity.

Take-up, claiming, and policy navigation

Intervention	Margin	Mechanism caution
Information letter	application or take-up	awareness versus salience
Reminder	filing or claiming timing	attention versus pressure
Simplified notice	completed claim	complexity versus hassle cost
Trusted messenger	program participation	information versus legitimacy

Bhargava-Manoli style designs identify treatment effects on claiming, but not a single mechanism without additional diagnostics.

Opaque incentives and workplace effort

- Firms write piece rates, bonuses, thresholds, penalties, and performance metrics.
- Workers may know there is an incentive while misunderstanding the formula.
- Observed margins: effort, output, task allocation, bonus attainment.
- Abeler-Huffman-Raymond style designs vary complexity and observe effort response.
- Workplace opacity can be accidental, productivity-enhancing, or strategic.

Savings, retirement, and training

- Payroll-linked saving depends on defaults, disclosures, and contribution understanding.
- Social Security and retirement claiming require mapping current work to future benefits.
- Training decisions require learning returns, subsidies, deadlines, and procedural steps.
- Observed margins: participation, contributions, claiming, enrollment, completion.
- Same diagnostic habit: what rule is perceived, what action moves, what remains confounded?

Methods and identification map

Design	Best identified object	What it cannot separate alone
Information letter or tutorial	missing information response	salience, trust, reminders
Simplification or disclosure	complexity and legibility	hassle costs, salience
Schedule change plus exposure	learning and attenuated response	sorting and diffusion
Monitored attention	costly information acquisition	experiment behavior
Workplace incentive experiment	effort under opacity	morale, peer effects, reciprocity
Structural learning model	welfare and counterfactuals	functional-form dependence

Welfare and design

- Stable inattention: observed choices may poorly reveal welfare.
- State-dependent inattention: workers may err after new rules, transitions, or shocks.
- Learning: short-run wedges and long-run wedges can differ sharply.
- Endogenous attention: optimal policy may reduce information costs rather than change payoffs.
- Design question: who makes the environment legible, and who benefits from opacity?

Week 4 lab

- Reproduce: synthetic Kostol-Myhre-style schedule-learning factbook.
- Diagnose: true incentives, perceived incentives, exposure, information, and hours response.
- Transfer: Bhargava-Manoli-style policy navigation with reminders and simplification.
- Optional extension: Abeler-Huffman-Raymond-style workplace incentive opacity.

Bridge to Week 5

- Week 4: workers respond to perceived and learned schedules.
- Week 5: firms design incentives, contracts, monitoring, and workplace communication.
- The bridge is contract legibility.
- Once workers are bounded, employer design becomes part of the labor mechanism.

Takeaways

- Attention, salience, complexity, and learning are dynamic labor mechanisms.
- The same reduced-form response can mix information, salience, hassle, optimization frictions, and learning.
- Labor applications must name the observed margin before naming the behavioral object.
- Welfare depends on persistence, heterogeneity, endogenous attention, and institutional design.